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VEHICLE DISPLAY

Abstract

A vehicle with a body defining a passenger compartment and having first and second support pillars on opposite sides on a front side of the passenger compartment. The vehicle has a windshield connected to the body and located between the first and second support pillars, the windshield having a windshield curvature. The vehicle has an instrument panel extending between the first and second support pillars. The vehicle has an electronic display substantially extending between the first and second support pillars and proximate to a lower portion of the windshield. The electronic display has a display curvature that is substantially similar to the windshield curvature.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] The present disclosure generally relates to a display arrangement on a vehicle, and more particularly relates to an electronic display installed proximate to the windshield on a motor vehicle.

BACKGROUND OF THE DISCLOSURE

[0002] Motor vehicles typically include electronic display devices for displaying information. For example, displays may be mounted proximate to the vehicle windshield, such as a Head-Up Display, or a display screen on the instrument panel or dashboard. It would be desirable to provide for other display arrangements for use on a motor vehicle.

SUMMARY OF THE DISCLOSURE

[0003] According to a first aspect of the present disclosure, a vehicle having a body defining a passenger compartment and having first and second support pillars on opposite sides on a front side of the passenger compartment, a windshield connected to the body and located between the first and second support pillars, the windshield having a configuration with a windshield curvature, a panel generally extending between the first and second support pillars below the windshield, and an electronic display substantially extending between the first and second support pillars and proximate to a lower portion of the windshield, wherein the electronic display has a display curvature that is substantially similar to the windshield curvature.

[0004] Embodiments of the first aspect of the present disclosure can include any one or a combination of the following features: [0005] the panel comprises an instrument panel; [0006] the mechanical support structure comprises a male member on one surface of the electronic display and the instrument panel configured to be received within a female receptacle on the other surface of the electronic display and the instrument panel; [0007] the electronic display is located vehicle rearward of the lower portion of the windshield; [0008] the mechanical support structure is configured to connect to an interior surface of the windshield; [0009] the windshield has the windshield curvature along the height of the windshield that substantially matches the display curvature along a height of the electronic display; [0010] the windshield curvature of the windshield along a width substantially matches the display curvature of the electronic display along the width; [0011] the electronic display is configurable as a split-screen display; [0012] the electronic display is configurable to have transparent and non-transparent portions; [0013] a mechanical support structure configured to connect the electronic display to the instrument panel; and [0014] the first and second support pillars extend vertically on opposite lateral sides of the body and are connected to a roof of the vehicle.

[0015] According to a second aspect of the present disclosure, a vehicle having a body defining a passenger compartment and having first and second support pillars, the windshield having a windshield curvature, a windshield connected to the body and located between the first and second support pillars, an instrument panel generally extending between the first and second support pillars below the windshield, an electronic display substantially extending between the first and second support pillars and proximate to a lower portion of the windshield, wherein the display has a display curvature that substantially matches the windshield curvature, and a support structure configured to connect the electronic display to the instrument panel.

[0016] Embodiments of the second aspect of the present disclosure can include any one or a combination of the following features: [0017] the mechanical support structure comprises a male member on one surface of the electronic display and the instrument panel configured to be received within a female receptacle on the other surface of the electronic display and the instrument panel; [0018] the male member is formed on a lower surface of the electronic display and the female

receptor is formed on a top surface of the instrument panel; [0019] the windshield has the windshield curvature along the height of the windshield that substantially matches the display curvature along a height of the display; [0020] the windshield curvature of the windshield along a width substantially matches the display curvature of the electronic display along the width; [0021] the electronic display is configurable as a split-screen display; [0022] the electronic display is configurable to have transparent and non-transparent portions; [0023] the electronic display is located vehicle rearward of the lower portion of the windshield; and [0024] the first and second support pillars extend vertically on opposite lateral sides of the body and are connected to a roof of the vehicle.
[0025] These and other features, advantages, and objects of the present disclosure will be further understood and appreciated by those skilled in the art by reference to the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0026] In the drawings:
[0027] FIG. **1** is a front view of the forward portion of an interior cabin of a motor vehicle having an electronic display mounted proximate to the vehicle windshield, according to one embodiment;
[0028] FIG. **2** is a side view of the vehicle windshield and the electronic display mounted on top of an instrument panel with a support structure, according to one example;
[0029] FIG. **3**A is a schematic view of the layers of the electronic display, according to a first example;
[0030] FIG. **3**B is a schematic view of the layers of the electronic display, according to a second example;
[0031] FIG. **3**C is a schematic view of the layers of the electronic display, according to a third example;
[0032] FIG. **4** is a side perspective view of the windshield and the electronic display located on the cabin interior side of the windshield;
[0033] FIG. **5** is a side view of the windshield and a support structure for mounting the electronic display to the windshield, according to another example;
[0034] FIG. **6** is a side view of the windshield and the electronic display having a support structure, according to a further example;
[0035] FIG. **7** is a front view of the windshield and the electronic display configured in a first operating mode;
[0036] FIG. **8** is a front view of the windshield and the electronic display configured in a split display function, according to a second operating mode; and
[0037] FIG. **9** is a front view of the windshield and the electronic display configured with transparent and non-transparent portions, according to a third operating mode.
DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS
[0038] Reference will now be made in detail to the present preferred embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numerals will be used throughout the drawings to refer to the same or like parts. In the drawings, the depicted structural elements are not to scale and certain components are enlarged relative to the other components for purposes of emphasis and understanding.
[0039] As required, detailed embodiments of the present disclosure are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the disclosure that may be embodied in various and alternative forms. The figures are not necessarily to a detailed design; some schematics may be exaggerated or minimized to show function overview. Therefore,

specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present disclosure.

[0040] For purposes of description herein, the terms "upper," "lower," "right," "left," "rear," "front," "vertical," "horizontal," and derivatives thereof shall relate to the concepts as oriented in FIG. **1**. However, it is to be understood that the concepts may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

[0041] The present illustrated embodiments reside primarily in combinations of method steps and apparatus components related to a vehicle having an electronic display. Accordingly, the apparatus components and method steps have been represented, where appropriate, by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein. Further, like numerals in the description and drawings represent like elements.

[0042] As used herein, the term "and/or," when used in a list of two or more items, means that any one of the listed items can be employed by itself, or any combination of two or more of the listed items, can be employed. For example, if a composition is described as containing components A, B, and/or C, the composition can contain A alone; B alone; C alone; A and B in combination; A and C in combination; B and C in combination; or A, B, and C in combination.

[0043] In this document, relational terms, such as first and second, top and bottom, and the like, are used solely to distinguish one entity or action from another entity or action, without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms "comprises," "comprising," or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by "comprises . . . a" does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.

[0044] As used herein, the term "about" means that amounts, sizes, formulations, parameters, and other quantities and characteristics are not and need not be exact, but may be approximate and/or larger or smaller, as desired, reflecting tolerances, conversion factors, rounding off, measurement error and the like, and other factors known to those of skill in the art. When the term "about" is used in describing a value or an end-point of a range, the disclosure should be understood to include the specific value or end-point referred to. Whether or not a numerical value or end-point of a range in the specification recites "about," the numerical value or end-point of a range is intended to include two embodiments: one modified by "about," and one not modified by "about." It will be further understood that the end-points of each of the ranges are significant both in relation to the other end-point, and independently of the other end-point.

[0045] The terms "substantial," "substantially," and variations thereof as used herein are intended to note that a described feature is equal or approximately equal to a value or description. For example, a "substantially planar" surface is intended to denote a surface that is planar or approximately planar. Moreover, "substantially" is intended to denote that two values are equal or approximately equal. In some embodiments, "substantially" may denote values within about 10% of each other, such as within about 5% of each other, or within about 2% of each other.

[0046] As used herein the terms "the," "a," or "an," mean "at least one," and should not be limited to "only one" unless explicitly indicated to the contrary. Thus, for example, reference to "a component” includes embodiments having two or more such components unless the context clearly indicates otherwise.

[0047] Referring to FIG. **1**, a motor vehicle **10**, such as a wheeled motor vehicle, is generally illustrated having a vehicle body **12** defining a cabin interior also referred to as a passenger compartment **14**. The passenger compartment **14** generally includes passenger seating, including a front row of seating having a driver seat **26**A and a passenger seat **26**B. The motor vehicle **10** may have one or more rear rows of seating. Located forward of the driver seat **26**A is a steering wheel **24**. A trim panel in the form of a dashboard, also referred to as an instrument panel **22**, extends forward of the first row of seating and forward of the steering wheel **24**.

[0048] The body **12** includes a windshield **20** generally extending across the front end of the passenger compartment **14**. The windshield **20** generally extends above and forward of the instrument panel **22**. The body **12** also includes a roof **16** along the top side and left and right side support pillars **18**A and **18**B shown located on the front lateral left and right sides of the vehicle body **12**. The support pillars **18**A and **18**B may extend from an underlying frame and connect to the roof **16** at the top end. The windshield **20** thereby extends between the left and right support pillars **18**A and **18**B on the left and right sides and the instrument panel **22** and roof **16** on the bottom and top sides.

[0049] The motor vehicle **10** includes an electronic display **30** shown substantially extending between the first and second support pillars **18**A and **18**B and proximate to a lower portion of the windshield **20** inside the passenger compartment **14**. The electronic display **30** is vehicle rearward of the windshield **20** and therefore located in the passenger compartment **14**. The electronic display **20** is positioned proximate the lower portion of the windshield **20** generally supported on top of the instrument panel **22** and extending substantially the full length between the first and second support pillars **18**A and **18**B. As such, the electronic display **30** is viewable by a driver and passenger looking forward in the motor vehicle **10** and viewing the electronic display **20** proximate the lower portion of the windshield **20**.

[0050] The electronic display **30** may be connected to the support pillars **18**A and **18**B on opposite lateral ends via fasteners and may further be connected to the instrument panel **22** and/or windshield **20**. The electronic display **30** may be mounted on top of the instrument panel **22** and held in place with a mechanical support structure **150** as shown in FIG. **2** in one example. In another example, the display may extend above the instrument panel **22** via a gap. The mechanical support structure **150** may include an angled support wall **156** engaging the inner surface on the lower portion of the windshield **20** and the top surface of the instrument panel **22**. The angled support wall **156** is on the vehicle forward side of the electronic display **30**. The angled support wall **156** may be curved to substantially match the curvature of the lower portion of the windshield **20** and may be connected to the instrument panel **22** and/or windshield **20** with fasteners and/or adhesive. In one embodiment, the width of the display **30** may be less than the distance between the first and second support pillars **18**A and **18**B.

[0051] The windshield **20** has a sloped contour with a windshield curvature along the width extending between the first and second support pillars **18**A and **18**B and along the height extending from the instrument panel **22** to the roof **16**. The electronic display **30** likewise has a sloped contour with a display curvature along the width and the height of the electronic display **30**. The display curvature substantially matches the windshield curvature. This matching curvature can be seen in FIG. **4** in which the windshield **20** curves from the bottom end on the top of the instrument panel **22** to the top end at the roof **16** and also curves along the width extending between the first and second support pillars **18**A and **18**B. The electronic display **30** likewise has a display curvature extending from the bottom end on the instrument panel **22** up until the proximate contact with the windshield **20** at the upper end and a display curvature substantially matching the windshield curvature extending along the width from the first support pillar **18**A to the second support pillar **18**B. As such, the windshield curvature of the windshield **20** substantially aligns with the display curvature of the electronic display **30**.

[0052] The electronic display **30** may be configured as a rigid display formed in the shape of the curved configuration. The electronic display **30** may be formed as a flexible display that may be flexed to assume the shape of the curved configuration when mounted onto the vehicle **10**. The electronic display **30** may include any of a number of electronic displays to display content such as the examples of displays illustrated in FIGS. **3**A-**3**C.

[0053] The electronic display **30** may be configured as an organic light emitting diode (OLED) display **30**A, according to one example shown in FIG. **3**A. The OLED display **30**A may be a flexible display having a TFT array **110** with an OLED layer **112** formed on top thereof. An encapsulation layer **114** may be formed on top of the OLED layer **112**. Finally, a polarizer **116** may be fabricated on top of the encapsulation layer **114**.

[0054] The electronic display **30** may be configured as a micro light emitting diode (LED) display **30**B as shown in the example in FIG. **3**B. The micro LED display **30**B likewise has a TFT array **210**. Disposed or layered on top of the TFT array **210** is an array of Red-Green-Blue (RGB) LEDs **212**. A polarizer layer **216** is formed on top of the LED layer **212**.

[0055] The electronic display **30** may be configured as a liquid crystal display (LCD) **30**C as shown in FIG. **3**C, according to a further example. The liquid crystal display **30**C employs a backlight layer **320** and a polarizer layer **318** disposed on top thereof and below a TFT array **310**. A liquid crystal layer **312** is disposed on top of the TFT array **310**. A color filter **314** is layered on top of the liquid crystal layer **312**. Finally, a polarizer **316** is shown layered on top of the color filter **314**. The liquid crystal display **30**C may employ a heat sink design for dissipating heat and generally may be a more rigid type of display compared to the other examples.

[0056] Referring to FIG. **4**, the electronic display **30** is generally illustrated having a configuration with a display curvature along the height or slope and the width of the motor vehicle **10**. Specifically, the electronic display **30** has a sloped curvature extending along the height that curves to match the sloped windshield of the windshield **20** extending between the lateral sides and along the height of the windshield **20**.

[0057] Referring to FIG. **5**, one example of a mechanical support structure **250** for mounting the electronic display **30** in the motor vehicle **10** is illustrated. In this example, the electronic display **30** is positioned supported on top of the top surface of the instrument panel **22** in close proximity and vehicle rearward of the windshield **20** along the lower portion thereof. The electronic display **30** is connected to the interior side of the windshield **20** by the mechanical support structure **250** having upper and lower support arms **40**A and **40**B. The upper support arm **40**A is connected to a first connector **42** on the inside surface of the windshield **20**. Likewise, the lower support arm **40**B is connected to a second connector **42**B on the interior surface of the windshield **20**. Each of the first and second connectors **42**A and **42**B may be connected to the interior side of the windshield **20** via an adhesive, for example. It should be appreciated that more than two support arms may be employed. As such, the electronic display **30** may be supported by the windshield **20** and the instrument panel **22** on the bottom side thereof.

[0058] Referring to FIG. **6**, the electronic display **30** is shown connected to the top surface of the instrument panel **20** by using a mechanical support structure **350** that includes on the front side a vertical male member **50** such as a cover glass or plastic lens having a rib or series of posts having an extended portion **56** extending downward on the bottom side of the electronic display **30**. The extended portion **56** of the male member **50** fits within a female receptacle **52** such as a slot or channel formed in the upper surface of the instrument panel **22** extending widthwise between the first and second support pillars **18**A and **18**B. As such, the lower portion of the post serves as a male member **50** extended within the female receptacle **52**. The male member **50** may be fastened or adhered to the female receptacle **52**. It should be appreciated that the mechanical support structure shown in FIG. **6** may have the male member **50** formed in one of the electronic display **30** and the instrument panel **20**, and the female receptacle **52** formed the other of the electronic display

30 and the instrument panel **20** such that the male member **50** and female receptacle **52** could be switched to be located on either component, according to another example.
[0059] Referring to FIGS. **7-9**, various examples of the electronic display **30** are illustrated configurable in different operating modes. In FIG. **7**, the electronic display **30** is shown configured to operate in a pillar-to-pillar display operating mode as a single display screen. In this mode, content including information such as text, video, icons, and other characters may be displayed across the entire length of the electronic display **30**.
[0060] The electronic display **30** may also be configured to operate in a split screen display mode, as seen in FIG. **8**. In this mode, the electronic display **30** may include a first display screen region as shown on the left half of the electronic display **30** and a second display screen region as shown on the right half of the electronic display. As such, different data may be displayed on each of the two split screen regions.
[0061] Referring to FIG. **9**, the electronic display **30** may be configured to operate to be split into a transparent display region and a non-transparent display region. The transparent display region is shown on the left side of the electronic display **30**, and the non-transparent display region is shown on the right side of the electronic display **30**. As such, a viewer located within the motor vehicle **10** may be presented with a transparent viewing area to look through the electronic display **30** and view the lower portion of the windshield **22** in the transparent portion but not in the non-transparent portion. The electronic display **30** may be configured to operate with more than one transparent display and more than one non-transparent regions.
[0062] Accordingly, the motor vehicle **10** advantageously employs an electronic display **30**, substantially extending between first and second support pillars **18**A and **18**B in the motor vehicle **10** proximate to a lower portion of the windshield **20** to provide an easy to view display. The electronic display **30** may be curved along the height and width that substantially matches the curvature of the windshield **20**. Additionally, the electronic display **30** may advantageously be mounted on top of the instrument panel **22** and may be connected to the instrument panel **22** via a mechanical support structure.
[0063] It is to be understood that variations and modifications can be made on the aforementioned structure without departing from the concepts of the present disclosure, and further it is to be understood that such concepts are intended to be covered by the following claims unless these claims by their language expressly state otherwise.

Claims

1. A vehicle comprising: a body defining a passenger compartment and having first and second support pillars on opposite sides on a front side of the passenger compartment; a windshield connected to the body and located between the first and second support pillars, the windshield having a configuration with a windshield curvature; a panel generally extending between the first and second support pillars below the windshield; and an electronic display substantially extending between the first and second support pillars and proximate to a lower portion of the windshield, wherein the electronic display has a display curvature that is substantially similar to the windshield curvature.
2. The vehicle of claim 1, wherein the panel comprises an instrument panel.
3. The vehicle of claim 2, wherein the mechanical support structure comprises a male member on one surface of the electronic display and the instrument panel configured to be received within a female receptacle on the other surface of the electronic display and the instrument panel.
4. The vehicle of claim 1, wherein the electronic display is located vehicle rearward of the lower portion of the windshield.
5. The vehicle of claim 2, wherein the mechanical support structure is configured to connect to an interior surface of the windshield.

6. The vehicle of claim 1, wherein the windshield has the windshield curvature along the height of the windshield that substantially matches the display curvature along a height of the electronic display.
7. The vehicle of claim 5, wherein the windshield curvature of the windshield along a width substantially matches the display curvature of the electronic display along the width.
8. The vehicle of claim 1, wherein the electronic display is configurable as a split-screen display.
9. The vehicle of claim 1, wherein the electronic display has transparent and non-transparent portions.
10. The vehicle of claim 2, further comprising a mechanical support structure configured to connect the electronic display to the instrument panel.
11. The vehicle of claim 1, wherein the first and second support pillars extend vertically on opposite lateral sides of the body and are connected to a roof of the vehicle.
12. A vehicle comprising: a body defining a passenger compartment and having first and second support pillars; a windshield connected to the body and located between the first and second support pillars, the windshield having a windshield curvature; an instrument panel generally extending between the first and second support pillars below the windshield; an electronic display substantially extending between the first and second support pillars and proximate to a lower portion of the windshield, wherein the display has a display curvature that substantially matches the windshield curvature; and a support structure configured to connect the electronic display to the instrument panel.
13. The vehicle of claim 12, wherein the support structure comprises a male member on one surface of the electronic display and the instrument panel configured to be received within a female receptacle on the other surface of the electronic display and the instrument panel.
14. The vehicle of claim 13, wherein the windshield has the windshield curvature along the height of the windshield that substantially matches the display curvature along a height of the display.
15. The vehicle of claim 12, wherein the windshield curvature of the windshield along a width substantially matches the display curvature of the electronic display along the width.
16. The vehicle of claim 15, wherein the curvature of the windshield substantially aligns to the curvature of the display.
17. The vehicle of claim 12, wherein the electronic display is configurable as a split-screen display.
18. The vehicle of claim 12, wherein the electronic display is configurable to have transparent and non-transparent portions.
19. The vehicle of claim 12, wherein the electronic display is located vehicle rearward of the lower portion of the windshield.
20. The vehicle of claim 12, wherein the first and second support pillars extend vertically on opposite lateral sides of the body and are connected to a roof of the vehicle.